

Proposal: BMW Car Sales Visualization Dashboard

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Problem Statement

BMW operates globally, offering a wide range of car models across different regions, fuel types, and transmissions. Understanding the **factors that influence sales volume and pricing** is essential for identifying market trends, customer preferences, and potential growth opportunities.

Hence, we need a dashboard that will serve as a **decision-support tool** for analyzing sales performance and can also be used for customer-facing exploration of BMW product trends.

Data Description

The dataset provided is the **BMW Car Sales Classification Dataset** with the following variables:

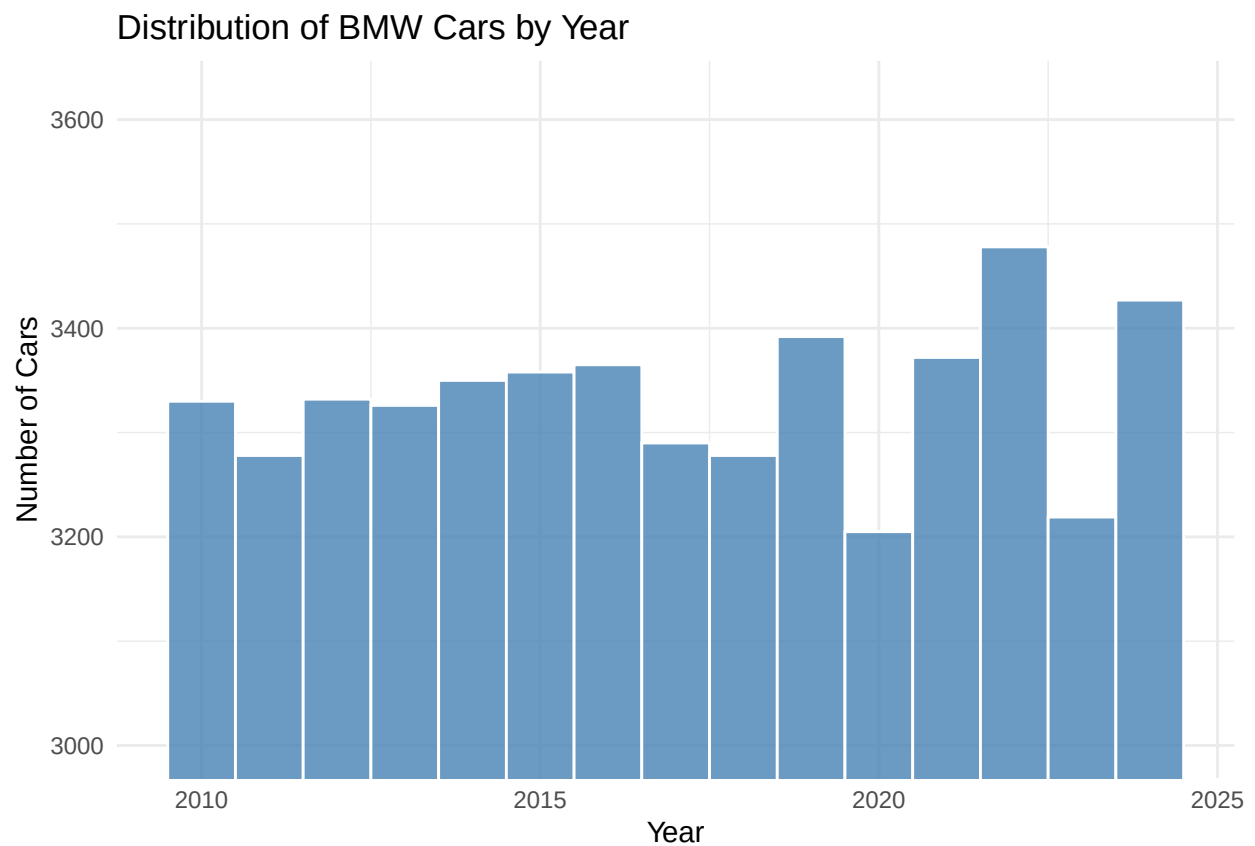
- **Model:** Specific BMW model name
 - **Year:** Manufacturing or sales year
 - **Region:** Geographic sales region
 - **Color:** Exterior color of the vehicle
 - **Fuel_Type:** Petrol, Diesel, Hybrid, or Electric
 - **Transmission:** Automatic or Manual
 - **Engine_Size_L:** Engine size in liters
 - **Mileage_KM:** Car mileage in kilometers
 - **Price_USD:** Sales price in US dollars
 - **Sales_Volume:** Number of units sold
 - **Sales_Classification:** Category (e.g., Low, Medium, High)
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Approaches

This data has many dimensions, and its nature offers a great variety. So I started with the question, ‘What all can I do with this data?’ First, I’m jotting down all the possible types of analysis and visualizations I could try. Then, I’ll pick the ones that work best together to represent the data clearly and simply, while also telling a compelling data story.

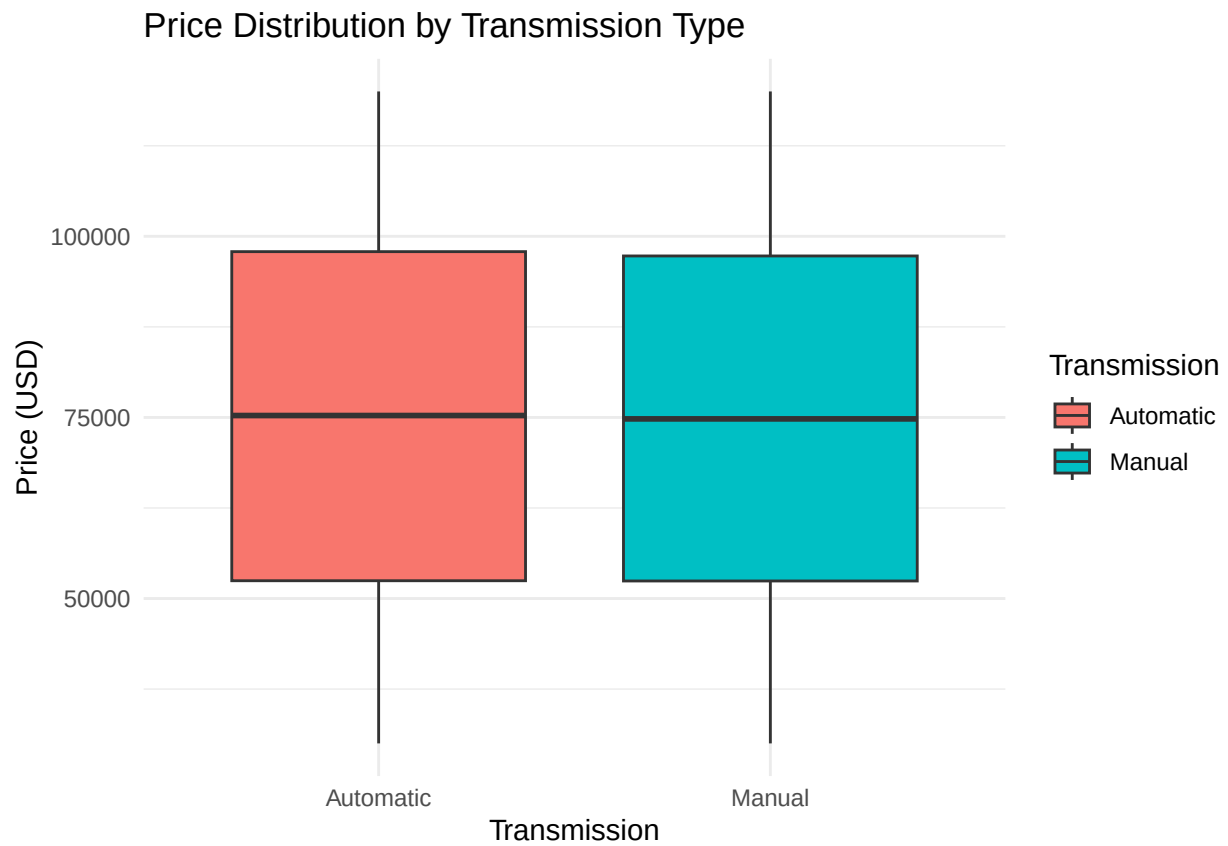
Univariate Analysis

- **Price distribution:** Histogram / density plot
- **Sales volume by region:** Bar chart
- **Fuel type share:** Donut chart



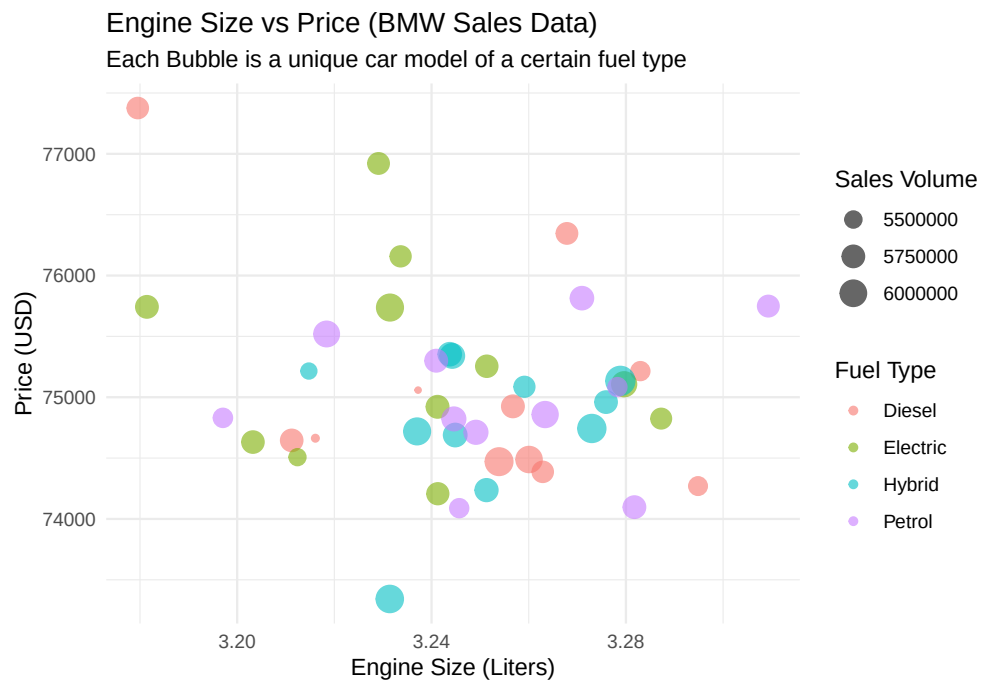
Bivariate Analysis

- **Price vs. Mileage:** Scatter plot
- **Price vs. Transmission / Fuel type:** Boxplots
- **Sales volume vs. Year:** Line chart



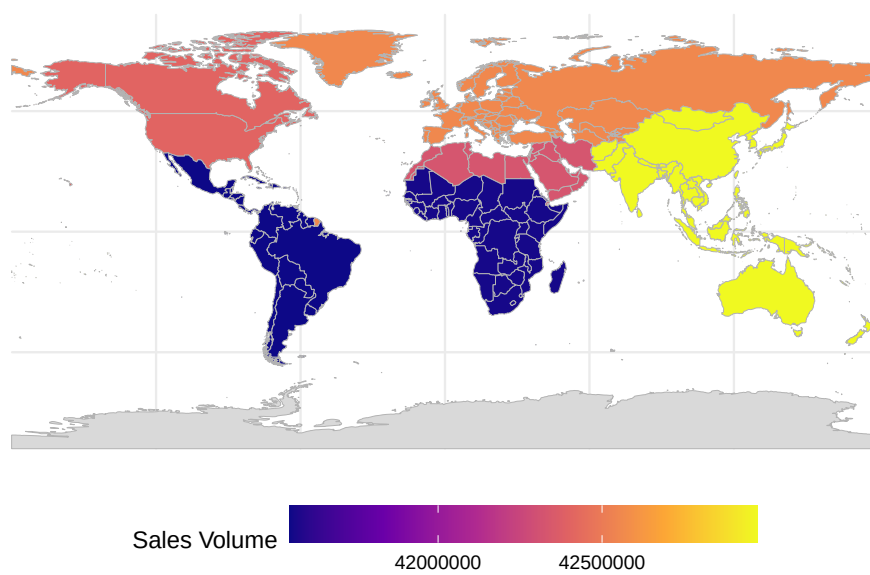
Multivariate Analysis

- **Bubble chart:** Engine size vs. Price, bubble size = Sales volume, color = Fuel type
- **Heatmap:** Region vs. Sales classification, intensity = sales volume
- **Geographical map:** World map with regions colored by sales volume



Example: BMW Sales by Region (Mock Data)

Darker = higher sales (sqrt scale), Gray = no data



Source: Mock Data

Dashboard Design

The Shiny dashboard will be designed with the following structure:

- **Sidebar Filters**
 - Model, Year, Region, Color, Fuel_Type, Transmission, Sales_Classification
- **Tabs**
 - *Overview*: Summary KPIs (average price, total sales, avg mileage) and a high-level trend plot
 - *Price Analysis*: Histograms, scatter plots, and boxplots
 - *Sales Analysis*: Line plots and bar charts of sales volume
 - *Advanced Exploration*: Bubble charts, heatmaps, and geographical maps

A good modern UI framework will be used with a color scheme inspired by BMW branding (blue, black, gray, white) to give the dashboard a visual touch while staying focused on the company color design principles.

Expected Outcome

The final deliverables will include:

1. A **Shiny dashboard** deployed online via shinyapps.io, enabling interactive exploration of BMW sales.
2. Insights into:
 - Regional differences in sales performance
 - The impact of mileage and engine size on pricing
 - Sales classification patterns across regions and fuel types
3. A clear and visually appealing interface that combines statistical insight with usability.